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Association of smoking exposure and sociodemographic factors with histological subtypes of bronchogenic carcinoma in upper Egypt

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Abstract

Background Lung cancer remains a leading cause of cancer-related mortality worldwide and poses a growing public health challenge in Egypt. Histopathological subtypes show varying associations with smoking behavior and sociodemographic factors, yet data from Upper Egypt remain scarce.

Objective To determine the association between smoking exposure, smoking-related characteristics, and sociodemographic factors with non-adenocarcinoma compared with adenocarcinoma among patients with bronchogenic carcinoma.

Patients and methods This case-case study was conducted at specialized oncology hospitals in Qena, Egypt, between May 2024 and October 2025, enrolling 150 patients: 50 with non-adenocarcinoma subtypes, including squamous cell carcinoma (SCC) and small-cell lung cancer (SCLC)—and 100 with adenocarcinoma. Sociodemographic and detailed smoking profiles were collected via structured face-to-face interviews, while clinical comorbidities and definitive histopathological classifications were extracted from validated medical records.

Results Among 150 patients (50 non-adenocarcinomas and 100 adenocarcinomas), non-adenocarcinoma was significantly more common in males (96% vs. 73%, $p < 0.001$) and current smokers (84% vs. 62%, $p < 0.001$). Patients with non-adenocarcinoma had substantially higher cumulative smoking exposure than those with adenocarcinoma [70.5 (49.1–78.7) vs. 40.0 (36.0–45.0) pack-years, $p < 0.001$]. In multivariable logistic regression, smoking exposure > 40 pack-years was strongly associated with non-adenocarcinoma histology (adjusted OR = 19.37; 95% CI: 3.45–108.72).

Conclusion This study provides crucial localized data from Egyptian patient population, demonstrating that higher cumulative smoking exposure is strongly associated with the non-adenocarcinoma histological subtype in bronchogenic carcinoma. Given these findings, targeted screening strategies and comprehensive smoking cessation programs in Egypt should prioritize patients exceeding these critical exposure thresholds to optimize early detection and intervention outcomes.

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Keywords Bronchogenic carcinoma, Adenocarcinoma, Non-adenocarcinoma, Smoking, Upper Egypt, Histopathology

Introduction

Lung cancer is a serious global health concern, profoundly affecting morbidity and mortality rates and ranking among the most prevalent cancers worldwide [1, 2]. Recently, lung cancer is projected to remain the most commonly diagnosed cancer in men globally and the second most common in women [3]. Approximately 1.8 million fatalities, or 18% of all cancer-related deaths, were attributed to lung cancer [4]. The five-year survival rate ranges from 10% to 20% [5]. It is the third most prevalent cancer among men in Egypt, accounting for 5–7% of diagnoses [6]. According to the most recent data, approximately 134,000 new cancer cases and over 30,000 cancer deaths were recorded annually in Egypt, with lung cancer contributing significantly to this burden [7]. Bronchogenic carcinoma is primarily categorized into two histological types: small cell lung cancer (SCLC) and non-small cell lung cancer (NSCLC) [8]. NSCLC, which constitutes approximately 85% of lung cancer cases, encompasses subtypes including adenocarcinoma and squamous cell carcinoma (SCC) [9]. Recently, the histological distribution of bronchogenic carcinoma has shifted, with adenocarcinoma exceeding SCC in prevalence [10].

Smoking is the primary risk factor for lung cancer, responsible for over 85% of global cases [11]. Smoking status strongly influences the histopathological subtypes of lung cancer; different subtypes show varying associations with tobacco exposure [12]. Smoking causes widespread carcinogenic changes in the bronchial epithelium that may influence tumor development and its distribution within the lungs [13]. Occupational diseases arise from long-term exposure to hazardous workplace factors, including physical, chemical, biological, ergonomic, and psychosocial agents. Occupational exposure to carcinogens, including asbestos, arsenic, respirable crystalline silica, chromium, diesel exhaust, nickel, beryllium, and cadmium, is significantly correlated with an elevated risk of bronchogenic carcinoma [14, 15]. In Egypt, occupational exposure to silica dust remains a significant public health concern in several industrial settings [16]. Environmental contaminants also play an important role in the development of bronchogenic carcinoma. Air pollution, nitrogen oxides, and industrial pollutants, induces oxidative stress, chronic airway inflammation, and genetic damage, potentially initiating carcinogenesis [17]. In addition, occupational and environmental exposures may interact synergistically with tobacco smoking, substantially increasing the overall risk of lung cancer [18].

Gender significantly influences the histopathological features of lung cancer, with noticeable differences in incidence and the most common histopathological subtypes between males and females [19]. In chronic obstructive pulmonary disease (COPD), increased levels of hypoxia-inducible factor 1-alpha (HIF-1α) resulting from chronic hypoxia facilitate angiogenesis and cellular survival, hence connecting inflammation and hypoxia to the progression of lung cancer [20, 21]. The association between lung cancer histological subtypes, gender, and smoking status in Egypt reflects a complex interplay of traditional tobacco exposure patterns, emerging environmental risk factors, and distinct molecular profiles. In the Egyptian context, NSCLC constitutes the predominant form of the disease, with adenocarcinoma and SCC representing the most frequent histological variants [22]. The distribution of these subtypes is heavily stratified by gender and smoking history, mirroring global trends but modulated by local epidemiological specificities such as the high prevalence of water-pipe smoking among men and significant exposure to biomass fuels and environmental tobacco smoke among women [23].

Lung cancer is a major health problem in Egypt. Yet, the impact of smoking and sociodemographic factors on its histopathology is understudied due to limited local data and poor statistical estimates, particularly in Upper Egypt [24]. So, further studies are needed to refine quantitative estimates of subtype distribution and to monitor temporal trends in lung cancer epidemiology, particularly in Upper Egypt. Accordingly, this study aimed to determine the association between histological subtypes of bronchogenic carcinoma (adenocarcinoma vs. non-adenocarcinoma) and smoking behavior and sociodemographic characteristics. Understanding these subtype-specific associations with smoking can inform targeted tobacco control policies in Egypt and similar settings.

Patients and methods

Study design and setting

A case-case study design was employed to compare clinical, behavioral, and sociodemographic profiles between different histological cohorts of lung cancer [25]. This study was conducted from May 2024 to October 2025 across three major oncology referral centers in the Qena Governorate: The Qena Oncology Center, the Oncology Department at Qena University Hospital, and the Health Insurance Hospital. These institutions collectively provide comprehensive diagnostic and therapeutic services to a diverse population of lung cancer patients originating

from various districts and socioeconomic backgrounds within Upper Egypt

The two comparison groups were

a) The primary case group consists of patients diagnosed with non-adenocarcinoma histological subtypes, including SCC and SCLC.
b) The alternative case group comprises patients diagnosed with lung adenocarcinoma.

Because the objective was to compare determinants of histological subtype rather than identify risk factors for lung cancer occurrence, a case-case design was considered appropriate. Detailed information regarding individual smoking behaviours and sociodemographic characteristics was systematically collected from both groups to evaluate and contrast the specific associations between tobacco exposure thresholds and individual histopathological subtypes.

Inclusion criteria

- Adult patients (> 18 years) with histopathological or cytological confirmation of primary lung cancer.
- Both newly diagnosed and previously diagnosed bronchogenic carcinoma patients who attended the participating hospitals during the study period.
- Patients were recruited from either the oncology inpatient wards or the outpatient clinics of the participating centres.
- Patients were willing to participate and provide informed consent.

Exclusion criteria

- Patients with metastatic involvement of the lung from an extrapulmonary primary malignancy.
- Patients with incomplete or unavailable data.

Sampling

Convenience sampling was used, which may limit generalizability due to potential selection bias

Sample size calculation

To calculate an adequate sample size for this case-case study, a standard statistical formula for comparing two independent proportions (representing two distinct diseased case groups) was utilized: [26]

$$n = \frac{\left(z\frac{\alpha}{2} + z\beta\right)^2 * p * (1-p) * (r+1)}{d^2 r}$$

n: number of non-adenocarcinoma cases.

Zα/2: This depends on the level of significance; for a 5% type I error ($P < 0.05$) and a 95% confidence interval, it is 1.96.

Zβ: This depends on the study's power; at 80%, it is 0.84.

r: The allocation ratio of the comparison groups = 2 (Two adenocarcinoma cases for every one non-adenocarcinoma case).

p = prevalence of smoking exposure among adenocarcinoma, equal to 52.6% [23].

p0 = risk of smoking exposure among adenocarcinoma cases = p = 0.526.

p1 = risk of smoking exposure among non-adenocarcinoma cases = $\frac{p0 \times OR}{1 + p0(OR-1)} = \frac{0.526 \times 3}{1 + 0.526 \times 2} = 0.769$

d = magnitude of difference we want to detect (p1-p0) = 0.769 – 0.526 = 0.243.

Odds Ratio (OR) = 3 [23]. So,

$$n = \frac{(1.96 + 0.84)^2 \times 0.526 \times (1 - 0.526) \times (2+1)}{0.243^2 \times 2} = 49.65$$

The result was 50 non-adenocarcinoma cases matched with 100 comparative adenocarcinoma cases (total = 150 patients).

Data collection techniques and tools

- Data were collected through face-to-face interviews using a questionnaire developed by the researchers after an extensive review of the relevant literature and study objectives. It included:
  - Sociodemographic data: Age, gender, residence, marital status, and occupation
  - Smoking behaviors: Smoking status, duration, pack-years, age of onset, cigarettes per day, and years since cessation of smoking.
- Some data were extracted from patients' medical files, including:
  - History of associated comorbidities: e.g., COPD, diabetes mellitus, hypertension, cardiac disease, chronic kidney disease (CKD), and Metabolic Dysfunction-Associated Steatotic Liver Disease (MASLD).
  - Histopathological subtypes of bronchogenic carcinoma: Adenocarcinoma, SCC, and SCLC were considered the main pathological subtypes.

Definitions

Smoking status: *Never smoker*: A person who has smoked less than 100 cigarettes in their life; A *current smoker* is someone who has smoked at least 100 cigarettes and smokes regularly or on some days; a former *smoker*: someone who has smoked at least 100 cigarettes but has quit [27]. The smoking index (pack-years) measures cumulative tobacco exposure, determined by multiplying the daily pack consumption by the total years of smoking [28].

Exposure variables

Smoking status, smoking intensity, pack-years, occupation, and relevant sociodemographic characteristics

Outcome variable

Histological subtypes of bronchogenic carcinoma

Ethical considerations

Before enrollment, all eligible participants were informed about the study's objectives, procedures, benefits, and potential risks. Informed consent was obtained from each participant before data collection. Participation was voluntary, and participants were assured that their information would remain confidential and would be used solely for research purposes. Participants were informed of their right to refuse participation or withdraw from the study at any time without affecting their medical care

Data management and statistical analysis

Data were collected, coded, revised, and analyzed using the Statistical Package for the Social Sciences (SPSS) version 29 (SPSS Inc., Chicago, IL, 2023). Categorical variables, presented as numbers and percentages, were compared using chi-square tests, with Cramer's V assessing effect size (small: 0.06 to <0.17, moderate: 0.17 to <0.29, large: ≥ 0.29). Quantitative variables were assessed for normality via Kolmogorov–Smirnov and Shapiro–Wilk tests. Normally distributed quantitative variables, reported as mean ± SD, were compared using independent t-tests, with Cohen's d for effect size (small: 0.2 to <0.5, moderate: 0.5 to <0.8, large: ≥ 0.8). Non-normally distributed variables were presented as median (interquartile range) and compared using the Mann-Whitney U test, with Rank-Biserial Correlation (r) assessing effect size (small: 0.10 to <0.30; moderate: 0.30 to <0.50; large: ≥ 0.50). Binary logistic regression identified predictors of non-adenocarcinoma (SCC and SCLC) versus adenocarcinoma. Smoking duration, cigarettes per day, age of smoking initiation, and gender were excluded from the multivariable regression model due to multicollinearity with pack-years. Multicollinearity was assessed using variance inflation factors (VIFs), and variables showing substantial multicollinearity with pack-years were excluded from the final model. Predicted probabilities were generated using the final multivariable logistic regression model, with the remaining covariates held constant. Statistical significance was defined as $p \leq 0.05$.

Results

Patients' ages ranged from 33 to 83 years, with a mean age of 61.97 ± 8.46 years. Most patients were male (80.7%), married (98%), and from rural areas (64.7%). Nearly half of the patients (51.3%) had non-manual occupations, while 24% were engaged in manual labor. There was a statistically significant difference in gender distribution between adenocarcinoma and non-adenocarcinoma groups; 73% of the adenocarcinoma group were male, compared with 96% in the non-adenocarcinoma group ($p<0.001$), with a moderate effect size (Cramer's V = 0.275). The non-adenocarcinoma group was older than the adenocarcinoma group (64.12 ± 7.16 vs. 60.89 ± 8.89 years, $p=0.027$). Comorbidities were more frequent in the non-adenocarcinoma group (92% vs. 76% in the adenocarcinoma group, $p=0.018$), particularly COPD (82% vs. 58%, $p=0.003$; moderate effect size). No significant differences were observed in residence, marital status, and occupation between the two groups ($p>0.05$) (Table 1).

The majority of the non-adenocarcinoma group were current smokers (84%), whereas never smokers comprised 31% of the adenocarcinoma group, with a moderate effect size ($p<0.001$, Cramer's V = 0.309). The two groups differed significantly in terms of pack-years of smoking and number of cigarettes per day ($p<0.001$; large effect size), duration of smoking ($p<0.001$; moderate effect size), and age of smoking initiation ($p<0.001$; small effect size). The pack-years of smoking were markedly higher among the non-adenocarcinoma group [70.50 (49.1–78.7)] than the adenocarcinoma group [40.0 (36.0–45.0)]. The non-adenocarcinoma group initiated smoking at a significantly younger age [18.5 (15.0–20.0) years] compared to the adenocarcinoma group [21.0 (17.0–23.0) years]. Years since smoking cessation showed no significant difference ($p=0.884$) (Table 2).

Right-sided lung cancer was more common in former (100%) and current (63.5%) smokers, whereas left-sided lung cancer was more common among never smokers (54.5%) ($p=0.002$, Cramer's V = 0.282). Right-sided lung cancer patients had a higher pack-year than left-sided lung cancer patients [46.0 (38.0–66.0) vs. 44.0 (39.0-53.9), $p=0.033$; moderate effect size]. Also, the duration of smoking showed a statistically significant difference ($p=0.021$), with a moderate effect size. However, the age of smoking initiation and the daily number of cigarettes showed no significant difference ($p>0.05$). While heavy smoking parameters were significantly associated with right-sided tumor laterality ($p<0.05$), the anatomical

Table 1 Sociodemographic features among adenocarcinoma and non-adenocarcinoma groups

Parameters			Overall (*n* = 150)	Non-adenocarcinoma (*n* = 50)	Adenocarcinoma (*n* = 100)	*p*-value	Effect size
			N (%)	*N* (%)	*N* (%)		
Gender ^{X2}	Male		121 (80.7%)	48 (96%)	73 (73%)	< 0.001*	0.275^{V}
	Female		29 (19.3%)	2 (4%)	27 (27%)		
Age groups ^{X2}	33-		13 (8.7%)	2 (4%)	11 (11%)	0.400	0.140^{V}
	51-		46 (30.7%)	14 (28%)	32 (32%)		
	61-		77 (51.3%)	28 (56%)	49 (49%)		
	≥ 71		14 (9.3%)	6 (12%)	8 (8%)		
Age (years) ^{t}	Mean ± SD		61.97 ± 8.46	64.12 ± 7.16	60.89 ± 8.89	0.027*	0.386^{d}
Residence ^{X2}	Urban		53 (35.3%)	22 (44%)	31 (31%)	0.116	0.128^{V}
	Rural		97 (64.7%)	28 (56%)	69 (69%)		
Marital status ^{X2}	Married		147 (98%)	50 (100%)	97 (97%)	0.216	0.101^{V}
	Non-married		3 (2%)	0 (0%)	3 (3%)		
Occupation ^{X2}	Non-manual laborer		77 (51.3%)	29 (58%)	48 (48%)	0.368	0.115^{V}
	Manual laborer	Aluminum factory	4 (2.7%)	3 (6%)	1 (1%)		
		Coal miners	5 (3.3%)	1 (2%)	4 (4%)		
		Quarry worker	1 (0.7%)	0 (0%)	1 (1%)		
		Farmer	26 (17.3%)	8 (16%)	18 (18%)		
	Unemployed		37 (24.7%)	9 (18%)	28 (28%)		
One or more Comorbidities			122 (81.3%)	46 (92%)	76 (76%)	0.018*	0.194^{V}
Hypertension			81 (54%)	24 (48%)	57 (57%)	0.297	0.085^{V}
COPD			99 (66%)	41 (82%)	58 (58%)	0.003*	0.239^{V}
Diabetes			44 (29.3%)	19 (38%)	25 (25%)	0.099	0.135^{V}
Cardiac disease			8 (5.3%)	5 (10%)	3 (3%)	0.072	0.147^{V}
MASLD			11 (7.3%)	2 (4%)	9 (9%)	0.268	0.090^{V}
Chronic kidney disease			4 (2.7%)	2 (4%)	2 (2%)	0.474	0.059^{V}

χ2:Chi-square test, *t* independent t-test, *V* Cramer's V, *d* Cohen's d, *SCC* squamous cell carcinoma, *SCLC* small cell lung cancer, *MASLD* Metabolic dysfunction-Associated Steatotic Liver Disease, *COPD* Chronic obstructive pulmonary disease

*Statistically significant at $p \leq 0.05$

Table 2 Smoking-related variables among adenocarcinoma and non-adenocarcinoma groups

Smoking parameters		Overall (*n* = 150)	Non-adenocarcinoma (*n* = 50)	Adenocarcinoma (*n* = 100)	*p*-value	Effect size
		N (%)	*N* (%)	*N* (%)		
Smoking status ^{X2}	Current smoker	104 (69.3%)	42 (84%)	62 (62%)	< 0.001*	0.309^{V}
	Former smoker	13 (8.7%)	6 (12%)	7 (7%)		
	Never smoker	33 (22%)	2 (4%)	31 (31%)		
		Median (IQR)	Median (IQR)	Median (IQR)		
Pack-years of smoking ^{MWU}		40.0 (30.0–50.0)	70.5 (49.1–78.7)	40.0 (36.0–45.0)	< 0.001*	0.657^{r}
Duration of smoking (years) ^{MWU}		40.0 (30.0–46.0)	44.5 (41.0–49.0)	40.0 (36.5–45.5)	< 0.001*	0.400^{r}
Age at starting smoking (years) ^{MWU}		20.0 (16.0-22.5)	18.5 (15.0–20.0)	21.0 (17.0–23.0)	< 0.001*	0.273^{r}
Cigarettes per day ^{MWU}		20.0 (17.3–25.0)	30.0 (21.3–37.5)	20.0 (18.7–20.0)	< 0.001*	0.646^{r}
Years since cessation of smoking ^{MWU}		5.0 (3.5-6.0)	4.5 (3.0-6.3)	5.0 (4.0–6.0)	0.884	0.012^{r}

χ2 Chi-square test, *V* Cramer's V, *MWU* Mann-Whitney U test, *r* Rank-Biserial Correlation

*Statistically significant at $p \leq 0.05$

distribution of the tumors did not differ significantly between the adenocarcinoma and non-adenocarcinoma case groups ($p = 0.811$) (Table 3).

Univariable analysis showed that current and former smoking, heavy smoking (≥ 20 cigarettes/day), and high cumulative exposure (> 40 pack-years) significantly increased the risk of non-adenocarcinoma compared to adenocarcinoma. Early smoking initiation, longer smoking duration, male sex, and COPD were also significant. Multivariable logistic regression identified 40 pack-years as a strong predictor of the non-adenocarcinoma group (AOR = 19.37; 95% CI: 3.45-108.72) (Table 4) (Fig. 1).

Table 3 Lung cancer laterality in relation to smoking-related variables and histological subtypes

Parameters		Overall	Laterality		*p*-value	Effect size
			Right-sided (*n* = 94)	Left-sided (*n* = 56)		
Smoking status[X2] N (%)	Current smoker	104	66 (63.5%)	38 (36.5%)	0.002*	0.282[V]
	Former smoker	13	13 (100%)	0 (0%)		
	Never smoker	33	15 (45.5%)	18 (54.5%)		
Histologic cell type[X2] N (%)	Non-adenocarcinoma	50	32 (64%)	18 (36%)	0.811	0.019[V]
	Adenocarcinoma	100	62 (62%)	38 (38%)		
Pack-years of smoking [MWU]		Median (IQR)	46.0 (38.0–66.0)	44.0 (39.0-53.9)	0.033*	0.174[r]
Duration of smoking (years) [MWU]		Median (IQR)	42.0 (40.0–48.0)	42.5 (36.5–47)	0.021*	0.189[r]
Age at starting smoking (years) [MWU]		Median (IQR)	20.0 (16.0–22.0)	20.5 (16.0–23.0)	0.549	0.049[r]
Cigarettes per day [MWU]		Median (IQR)	20.0 (20.0–30.0)	20.0 (18.9–30.0)	0.060	0.154[r]

χ2 Chi-square test, *t* independent t-test, *V* Cramer's V, *MWU* Mann-Whitney U test, *r* Rank-Biserial Correlation

*Statistically significant at $p \leq 0.05$

Table 4 Univariable and multivariable logistic regression analysis of predictors for the non-adenocarcinoma group compared to the adenocarcinoma group

Predictor variable		Univariable analysis		Multivariable analysis	
		UOR (95% CI)	*p*-value	AOR (95% CI)	*p*-value
Smoking status	Never smoker	1			
	Current smoker	20.83 (2.74-158.38)	0.003*		
	Former smoker	27.43 (2.84-265.35)	0.004*		
Cigarettes per day	Never smoker	1			
	< 20	1.72 (0.14–21.24)	0.672		
	≥ 20	12.35 (2.81–54.26)	< 0.001*		
Pack-years of smoking	Never smoker	1			
	≤ 40	1.55 (0.27–9.02)	0.626	1.20 (0.17–8.59)	0.854
	> 40	24.91 (5.53-112.28)	< 0.001*	19.37 (3.45-108.72)	< 0.001*
Age at smoking initiation	Never smoker	1			
	20–29 years	12.80 (1.63-100.84)	0.015*		
	< 20 years	35.84 (4.56-281.79)	< 0.001*		
Duration of smoking	Never smoker	1			
	30–39 years	10.18 (1.17–88.68)	0.036*		
	≥ 40 years	27.83 (3.64-212.92)	0.001*		
Gender	Male	17.78 (2.34-135.12)	0.005*		
COPD		3.64 (1.55–8.57)	0.003*	1.28 (0.34–4.83)	0.720

COPD Chronic obstructive pulmonary disease, *UOR* Unadjusted odds ratio, *AOR* Adjusted odds ratio, *CI* Confidence interval

*Statistically significant at $p \leq 0.05$

Discussion

This case-case study was conducted across specialized oncology hospitals in Qena, Egypt, between May 2024 and October 2025. The study cohort comprised 150 total patients, partitioned into a primary case group of 50 non-adenocarcinoma patients and a comparative case group of 100 adenocarcinoma patients. The mean age of patients was 61.97 ± 8.46 years, and most patients were male (80.7%), married (98.0%), and residing in rural areas (64.7%). The observed gender differences, with non-adenocarcinoma subtypes occurring almost exclusively among males, likely reflect disparities in smoking prevalence and intensity in the study population. This aligns with a 2020 study in Eastern Anatolia by Özmen & Ceylan, who reported a higher prevalence of SCC and SCLC among males and adenocarcinoma among females [29]. In Upper Egypt, smoking remains substantially more common among men, which may contribute to both higher lung cancer incidence and a predominance of smoking-related histological subtypes in this group. However, the presence of adenocarcinoma among never smokers, particularly females, highlights the need to consider alternative risk factors such as indoor air pollution, occupational exposures, and passive smoking.

Age-related differences across histological subtypes were also identified, with the non-adenocarcinoma group being older than the adenocarcinoma group (64.12 ± 7.16 vs. 60.89 ± 8.89 years, $p = 0.027$). This finding aligns with studies from China in 2021 by Zou et al. and the National Cancer Institute's Surveillance, Epidemiology, and End

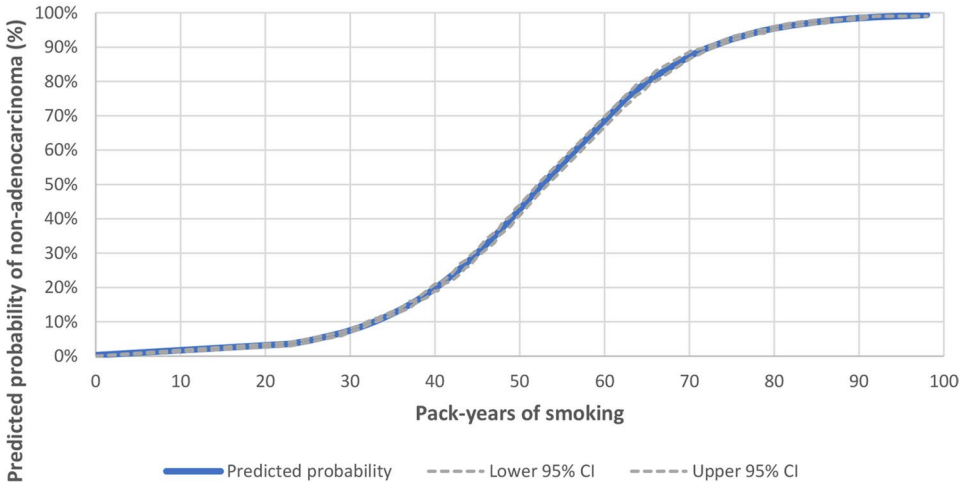


Fig. 1 Predicted probability of non-adenocarcinoma subtypes according to pack-years of smoking

Results (SEER) database in 2024 by Cittolin-Santos et al.., showing an increasing incidence of SCLC with age [30, 31]. This may reflect the cumulative effect of prolonged tobacco exposure, as well as age-related susceptibility to aggressive tumor phenotypes.

Current smokers comprised 62% of the adenocarcinoma group, compared to 84% of the non-adenocarcinoma group. Never smokers accounted for 31% of the adenocarcinoma group, but only 4% of the non-adenocarcinoma group. This agrees with previous studies in the US (Kuśnierczyk, 2023) and China Zhao et al., 2025), showing SCC is more frequent in heavy smokers, while adenocarcinoma is increasingly seen in non-smokers [32, 33]. Smoking remains the primary etiologic factor for lung cancer in Egypt, with a strong correlation observed between cumulative tobacco exposure and specific histological subtypes. SCC and SCLC are predominantly associated with active tobacco smoking, particularly in males who exhibit higher rates of cigarette consumption [34]. Conversely, adenocarcinoma is increasingly recognized as the most common subtype among never-smokers and females [23].

The relatively higher proportion of adenocarcinoma observed among females and never-smokers than non-adenocarcinoma in the present study can be explained by three interrelated factors: the physical deposition characteristics of environmental carcinogens, sex-specific anatomical differences, and distinct molecular drivers. In never-smokers, exposure to carcinogens occurs primarily through secondhand smoke, residential radon, ambient air pollution, and household coal use [35]. These sources predominantly emit fine PM2.5 and radioactive decay products (in the case of radon) that naturally possess small aerodynamic diameters. Due to their size, these particles inherently penetrate deep into the peripheral lung regions regardless of inhalation depth, although deeper breaths can enhance this effect. Consequently, the distal lung parenchyma is the primary site of carcinogen accumulation in non-smokers, predisposing them to adenocarcinoma rather than central SCC. Women exhibit specific anatomical and physiological traits that increase their vulnerability to peripheral lung cancer. On average, women have smaller lung volumes and narrower airway calibers compared to men [36]. For a given tidal volume or inhalation pattern, these anatomical constraints can result in a higher fraction of inhaled particles reaching the peripheral alveoli, thereby increasing the dose of carcinogens to the distal lung tissue.

Perhaps the most critical factor explaining the predominance of adenocarcinoma in women and never-smokers is the distinct molecular profile of their tumors. Lung cancers in never-smokers and women are far more likely to harbor specific oncogenic driver mutations, particularly in the epidermal growth factor receptor (EGFR) gene, as well as alterations in ALK, ROS1, and KRAS [37]. These driver mutations are overwhelmingly associated with the adenocarcinoma histological subtype. For example, EGFR mutations are found in a significant proportion of lung adenocarcinomas in never-smokers and women, whereas they are rare in SCC and SCLC. This suggests that the biological pathway to cancer in these populations is fundamentally different, driven by genetic susceptibilities and specific carcinogenic interactions that favor the development of adenocarcinoma in the peripheral lung, independent of the heavy central airway irritation that characterizes SCC in heavy smokers.

A significant association between smoking intensity and histopathological subtype was observed. The non-adenocarcinoma group had markedly higher pack-years, earlier smoking initiation, and longer smoking duration compared to those with adenocarcinoma. Similarly, in Bangladesh, Hassan et al., revealed that SCC and SCLC

patients began smoking earlier (22.1 ± 3.9 vs. 26 ± 3.7 years, $p < 0.001$), smoked longer (34.5 ± 8.9 vs. 18.5 ± 8.5 years, $p < 0.001$), and had greater pack-years (48.1 ± 12.5 vs. 12.8 ± 11, $p < 0.001$) compared to adenocarcinoma patients [10]. These findings support a dose–response relationship between tobacco exposure and central airway malignancies. Biologically, tobacco carcinogens induce widespread genetic mutations and epithelial damage, particularly in the central bronchi, favoring the development of SCC and SCLC. In contrast, adenocarcinoma appears less tightly linked to heavy smoking and may involve additional mechanisms, including genetic susceptibility, environmental exposures, and hormonal influences.

Additionally, the higher prevalence of COPD among the non-adenocarcinoma group aligns with an Egyptian study from Al-Azhar University, Cairo, which reported higher COPD prevalence in SCLC (85.7%) and SCC (61.5%) and lower prevalence in adenocarcinoma (28.9%) [23]. This supports the hypothesis that chronic inflammation and hypoxia contribute to carcinogenesis. The role of hypoxia-inducible pathways, including HIF-1α activation, may further link chronic lung disease with tumor progression and angiogenesis.

The prognosis and mortality associated with lung cancer are profoundly influenced by its histological subtype, which dictates biological behavior, therapeutic responsiveness, and overall survival outcomes. The prognosis of adenocarcinoma is heterogeneous, largely driven by its molecular and architectural diversity [38]. Histologic grading based on predominant patterns—such as lepidic, acinar, papillary, solid, or micropapillary further refines prognosis, with high-grade patterns like solid and micropapillary associated with poorer outcomes and higher recurrence rates [39]. In contrast, SCC typically exhibits slower growth and later metastasis compared to other subtypes, yet its prognosis remains poor due to limited targeted therapeutic options [40]. SCLC represents a highly aggressive neuroendocrine tumor characterized by rapid proliferation, early metastasis, and a high initial response to chemotherapy followed by inevitable relapse [41]. SCLC has the worst prognosis among lung cancer subtypes, with median survival often measured in months for extensive-stage disease [42].

An observed finding in this study was the predominance of right-sided lung cancer among smokers, which aligns with a Brazilian study in 2006, which noted that bronchogenic carcinoma in smokers was more frequently seen in the right lung compared to non-smokers [43]. Although the underlying mechanism remains unclear, it has been suggested that anatomical differences in bronchial structure and airflow dynamics may influence the distribution of inhaled carcinogens. It might be related to right bronchus geometry (which is wider, shorter, and more vertical, leading to preferential deposition of particulates). This observation warrants further investigation but may have implications for diagnostic vigilance in high-risk populations. The current study reveals a significant association between cumulative smoking exposure and tumor laterality in lung cancer patients, specifically indicating that right-sided tumors are associated with higher pack-year histories and longer smoking durations. This finding suggests that the anatomical distribution of lung cancer may be influenced by the intensity and duration of tobacco exposure, potentially reflecting differential deposition patterns of carcinogens or distinct biological responses in the right versus left lung. This observation is consistent with a recent meta-analysis by Ai et al., which demonstrated a significant dose–response relationship between active smoking exposure and both lung cancer prevalence and mortality, with progressively increasing risks among heavier smokers [44].

From a public health perspective, the strong association between smoking exposure and non-adenocarcinoma lung cancer underscores the importance of comprehensive tobacco control strategies, including smoking prevention, smoking cessation services, enforcement of smoke-free policies, and public awareness campaigns. Furthermore, identification of high-risk groups based on demographic and exposure characteristics may support targeted screening and early detection efforts. Future community-based studies are warranted to evaluate the effectiveness of such interventions in reducing lung cancer incidence and mortality in Egypt.

This study has several strengths, including detailed smoking-exposure metrics and effect-size measures, which enhance the interpretability and clinical relevance of the findings. However, the results should be viewed with some limitations in mind. Because this is a hospital-based case-case study, causal relationships cannot be established, and temporal relationships between smoking exposure and histological subtype should be interpreted with caution. In addition, convenience sampling from three referral hospitals may have introduced selection bias and limited the generalizability of the findings. Recall bias due to self-reported smoking history. Residual confounding from unmeasured variables such as air pollution, biomass fuel exposure, and molecular alterations (e.g., EGFR and ALK mutations) cannot be excluded. Despite these limitations, this study contributes valuable evidence from a region with limited epidemiological data. It highlights the heterogeneity of lung cancer subtypes and reinforces the central role of smoking exposure in shaping disease patterns. Future research should incorporate longitudinal designs, larger multicenter samples, and a more comprehensive assessment of environmental and genetic factors to better understand subtype-specific risk profiles.

Conclusion and recommendation

Our findings provide essential insights specifically derived from an Egyptian patient base. Adenocarcinoma was more frequently observed among females and never smokers than non-adenocarcinoma subtypes. In contrast, non-adenocarcinoma subtypes (SCC and SCLC) were strongly associated with male sex, current smoking, early initiation, and high cumulative tobacco exposure. These findings emphasize the importance of targeted tobacco control strategies and early prevention efforts to reduce the burden of aggressive lung cancer subtypes in Egypt and similar settings. Identifying challenges by histological subtypes aligns with global efforts to refine health estimates and inform intervention planning.

Abbreviations

CI	Confidence interval
COPD	Chronic obstructive pulmonary disease
CKD	Chronic kidney disease
EGFR	Epidermal growth factor receptor
HIF-1α	Hypoxia-inducible factor 1-alpha
IQR	Interquartile range
MASLD	Metabolic dysfunction-Associated Steatotic Liver Disease
NSCLC	Non-small-cell lung cancer
OR	Odds ratio
SCC	Squamous cell carcinoma
SCLC	Small-cell lung cancer
SEER	Surveillance, Epidemiology, and End Results
SPSS	Statistical Package for the Social Sciences
VIF	Variance inflation factors

Supplementary Information

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Supplementary Material 1.

Authors' contributions

First author: Data collection, statistical analysis, and manuscript drafting; Second author: Study conception and design, literature review, and critical revision of the manuscript; Third author: Literature review and critical revision of the manuscript; Fourth author: Manuscript writing and critical revision of the manuscript. All authors read and approved the final version of the manuscript and agreed to be accountable for all aspects of the work.

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Data availability

The datasets generated and/or analyzed during the current study are not publicly available because they contain information that could compromise participant privacy, but are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The study was approved by the Ethics Committee of Qena Faculty of Medicine in February 2024. The ethical approval code: SVU-MED-COM009-2-24-2-809. Informed consent was obtained from all participants before enrollment.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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